

Challenge (Habanero)

- Q1.** Simplify $2^3 \cdot 2^4$
(A) 2^{3+4}
(B) $2^{3 \cdot 4}$
(C) 2^{3-4}
(D) 2^{4-3}
(E) $2^{3/4}$
- Q2.** A recipe calls for 3^2 cups of flour. If the recipe is quadrupled, how many cups are needed?
(A) 3^2
(B) $4 \cdot 3^2$
(C) $3^2 \cdot 3^2$
(D) $3^{2 \cdot 4}$
(E) 3^{2+4}
- Q3.** A building is 2^5 meters tall. If a new building is 2^3 times taller, how tall is the new building?
(A) 2^5
(B) 2^8
(C) 2^{5+3}
(D) 2^{5-3}
(E) $2^{5 \cdot 3}$
- Q4.** Simplify $(2^2)^3$
(A) 2^{2+3}
(B) 2^{2-3}
(C) $2^{2 \cdot 3}$
(D) $2^{3/2}$
(E) $2^{2/3}$
- Q5.** A cake recipe requires 2^4 eggs. If the recipe is halved, how many eggs are needed?
(A) 2^4
(B) 2^3
(C) 2^{4-1}
(D) $2^{4/2}$
(E) 2^{4-2}
- Q6.** Simplify $3^2 \div 3^4$
(A) 3^{2-4}
(B) 3^{2+4}
(C) $3^{2 \cdot 4}$
(D) 3^{4-2}
(E) $3^{4/2}$
- Q7.** A water tank can hold 2^6 liters. If 2^4 liters are already filled, what fraction of the tank is filled?
(A) 2^{6-4}
(B) 2^{4-6}
(C) $2^{4/6}$
(D) $2^{6/4}$
(E) $2^6 \div 2^4$
- Q8.** Simplify $2^3 \cdot 2^2 \div 2^2$
(A) 2^{3+2-2}
(B) 2^{3-2+2}
(C) $2^{3 \cdot 2 \div 2}$
(D) $2^{3 \div 2 \cdot 2}$
(E) $2^{3 \cdot 2}$

Open Ended Questions (FRQ)

- Q9.** A construction project requires 3^4 bags of cement. If each bag weighs 2^3 kg, what is the total weight of cement needed? Show your work.
- Q10.** A recipe for making cookies calls for 2^5 cups of sugar. If the recipe is tripled, and then the amount of sugar is reduced by 2^2 cups, how many cups of sugar are needed? Explain your reasoning.

Chef's Tips

- Make sure to check the units of the answers
- Use the exponent rules to simplify expressions
- Check for any calculation errors